Ontology-Guided Screening for Undetected Dysglycemia: Evaluating Clinical Knowledge Integration Under Temporal Shift in NHANES

ClinOntoScreen Research Prototype — Manuscript Draft

Notice: Research prototype only. Not a diagnostic tool, not clinically validated, and not for medical decisions.

Abstract

Background. Clinical guidelines and tabular machine learning are usually developed and evaluated separately; few studies test whether encoding guideline content as executable, provenance-carrying knowledge changes downstream screening decisions, and almost none do so under a temporal shift that can break hard-coded thresholds.

Methods. We built ClinOntoScreen: an OWL/RDF ontology encoding ADA 2026 diabetes-screening recommendations as executable rules with guideline-section references, coupled to logistic regression and XGBoost models. Leakage-safe predictors (excluding HbA1c, glucose, questionnaire answers, and medications) were derived from NHANES 2017–March 2020 (historical development, n = 6,103, 28.9% screen-positive) and August 2021–August 2023 (locked temporal test, n = 4,260, 23.1% screen-positive). The outcome is HbA1c ≥ 5.7% among adults reporting neither prior diabetes nor prior prediabetes. The primary confirmatory endpoint was sensitivity of a hybrid ranking (XGBoost + ontology rule score) minus XGBoost alone at a fixed 20% screening capacity (paired bootstrap, 1,000 replicates). Exploratory experiments probed low-data regimes (E1), masked inputs (E2), model families (E3), disagreement review (H3), and the published CDC/ADA risk test.

Results. The primary effect was null: hybrid 0.409 vs ML-only 0.412 at 20% capacity; in the prespecified direction (hybrid − ML-only) the difference was **−0.003 (95% CI −0.010 to +0.008)**, entirely within the smallest detectable difference (0.014). At 20% capacity the prevalence ceiling is 0.867; the best policy (logistic + ontology features) reached sensitivity 0.423 (0.488 of ceiling), PPV 0.488, lift 2.12. All trained models outperformed the published CDC/ADA score (AUPRC 0.460–0.463 vs 0.380). Ontology features helped the linear model consistently in direction (sensitivity@20% +0.009, CI −0.004 to +0.019) and gave the smallest degradation under 30% input masking, but no interval excluded zero.

Conclusions. Encoding guideline provenance into an executable layer is feasible and auditable, but on NHANES with full historical data the rule-derived features are largely redundant with raw inputs for ranking. The honest headline is a well-powered null on the primary endpoint, with a consistent advantage for guideline features in linear, low-data, and missing-input settings.

Keywords: clinical ontology, screening, NHANES, temporal validation, fixed-capacity evaluation

1. Introduction

Diabetes screening is a resource-allocation problem before it is a prediction problem: practices can screen only a bounded fraction of their population, so a model's value at a fixed referral capacity is often more decision-relevant than its AUROC. At the same time, guidelines such as the ADA Standards of Care encode expert consensus that is auditable in prose but inert in software. Tabular machine learning can rank risk

effectively, yet it silently discards that consensus; guideline rule engines keep the consensus but usually are not evaluated under strict temporal generalization tests.

ClinOntoScreen asks a narrow, testable question: **does encoding a documented subset of screening guidance as an executable, provenance-carrying ontology change screening decisions or robustness, relative to raw features, when both are evaluated on a genuinely later survey cycle?**

Key Contributions:

• An OWL/RDF ontology whose executable rules carry guideline section references and operationalization status.

• A leakage-safe, cycle-harmonized feature contract with automated contract testing.

• A capacity-constrained evaluation protocol with prevalence ceilings and pre-registered MDE thresholds.

• An honest temporal evaluation on the latest cycle (2021–2023) against published knowledge and stress probes.

2. Related Work

NHANES supports a large prediction literature; closest to us in evaluation discipline is Kumar's leakage-controlled benchmark with rolling-origin temporal cross-validation and the same 2021–2023 holdout. Pham and Soo combine community detection with a diabetes ontology on NHANES data and report high discrimination, but with lab-inclusive features and a random split. Random splits overstate performance under survey-cycle shift; our ceiling-adjusted capacity reporting makes the achievable operating point explicit.

Table 1: Capability comparison with related studies.

Dimension	Pham & Soo (CEAI)	Kumar (CMPB)	ClinOntoScreen (This work)
Data	Pima, Hospitals, NHANES	NHANES dev + holdouts	NHANES dev + 2021–2023 temporal test
Target	Diabetes (lab-inclusive)	HbA1c ≥ 6.5%	HbA1c ≥ 5.7% (no prior history)
Knowledge Role	Domain ontology + clustering	None	OWL/RDF rules with provenance
Primary Metric	Accuracy / AUROC	Discrimination + calibration	**Sensitivity @ capacity with ceiling**

3. Data and Cohort Harmonization

Sources & Cohort Flow. CDC/NCHS public XPT files across NHANES cycles. Adults aged 18+ with valid HbA1c and no reported prior diabetes or prediabetes comprise the historical development cohort (2017–March 2020, n = 6,103, 28.94% screen-positive) and the temporal test cohort (August 2021–August 2023, n = 4,260, 23.08% screen-positive).

Covariate Shift & Harmonization. The 2021–2023 test cycle is older (mean age 49.2 vs 46.1) and demographically distinct. Physical activity instruments were harmonized across incompatible survey instruments into leisure minutes and sedentary minutes. HDL cholesterol is included; hypertension treatment

(`BPQ040A`) and family history (`MCQ300C`) exist only in the earlier cycle (disclosed as 100% missing in test). Survey weights were not applied in this study.

4. Ontology and Rule Engine

The OWL/RDF knowledge model (version 0.2.0) encodes screening rules, age thresholds, overweight/obesity criteria (25 kg/m² general; 23 kg/m² for Asian participants), hypertension, physical inactivity, CVD history, and higher-risk ethnicity. Every rule carries guideline section references (`sectionReference`) and operationalization status. The Python rule executor derives 11 `onto_*` features and produces a rule-priority score.

5. Experimental Methods

Leakage Contract. Predictors strictly exclude HbA1c, glucose, diabetes questionnaire responses, medications, and IDs. Preprocessing pipelines fit exclusively within each training fold.
Models & Policies. L2 logistic regression and XGBoost. Hyperparameters tuned on an 80/20 historical split (n = 1,221). Six model variants evaluated. Policies include raw probability, hybrid score (alpha = 0.10), rule-only score, and published CDC/ADA risk test.
Evaluation. Test outcomes read once after protocol freezing. Metrics include AUROC, AUPRC, Brier score, and capacity tables at 10%, 20%, 30% referral budgets reporting prevalence ceiling, fraction of ceiling, PPV, and lift. Uncertainty evaluated via paired participant bootstrap (1,000 resamples) and pre-registered MDE thresholds.

6. Empirical Results

6.1 Historical Validation

Historical validation demonstrated robust baseline performance across models, with AUROC ranging from 0.776 to 0.781 and AUPRC from 0.542 to 0.549. Hybrid alpha was selected as 0.10.

6.2 Temporal Test Discrimination & Calibration

On the locked 2021–2023 temporal test set, trained models achieved strong discrimination despite distribution shift.

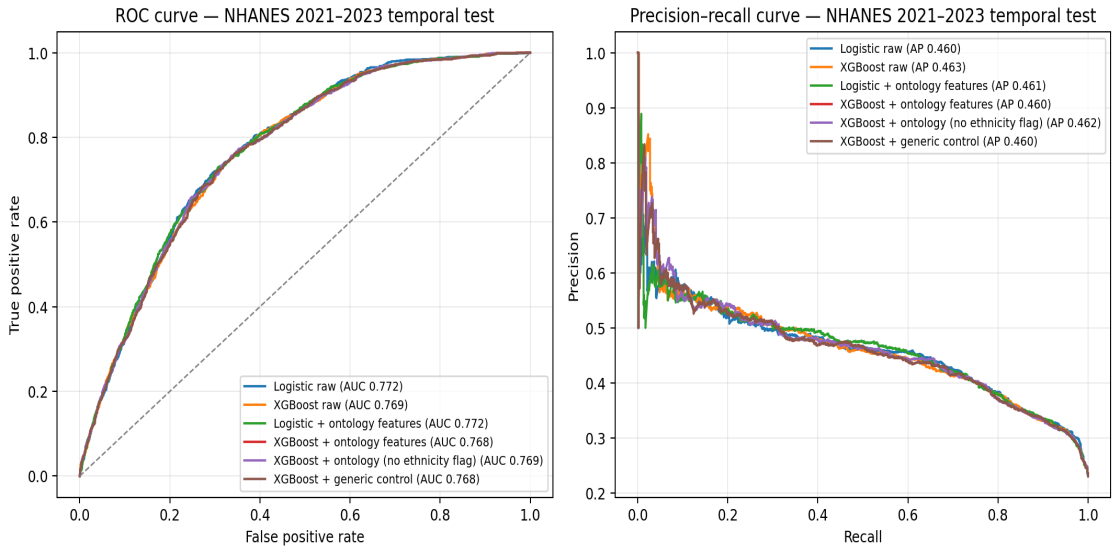


Figure 1a: *Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves.*

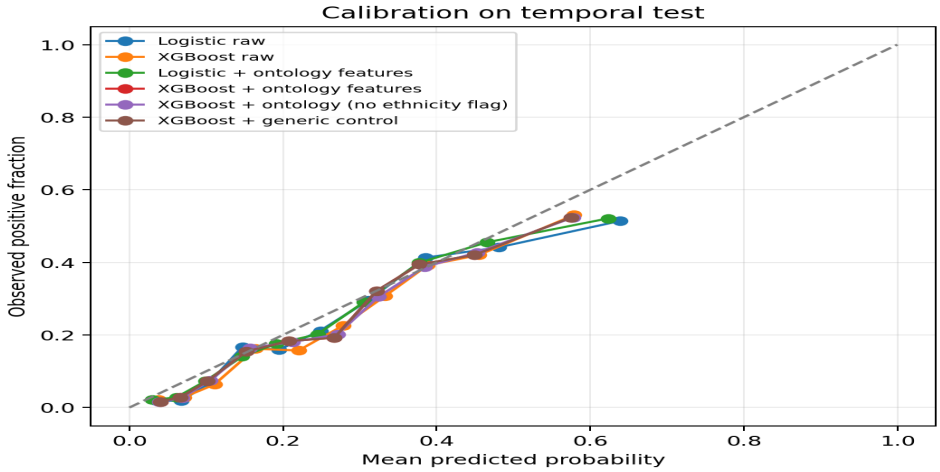


Figure 1b: *Calibration reliability curves on the temporal test set.*

6.3 Fixed Screening Capacity

Evaluating policies at fixed screening capacities against the prevalence ceiling (0.867 at 23.08% prevalence) shows that logistic regression combined with ontology features achieves the highest sensitivity at 20% capacity (0.423, finding 416 of 983 positives) with a PPV of 0.488 and lift of 2.12. All trained models outperformed the published CDC/ADA risk test.

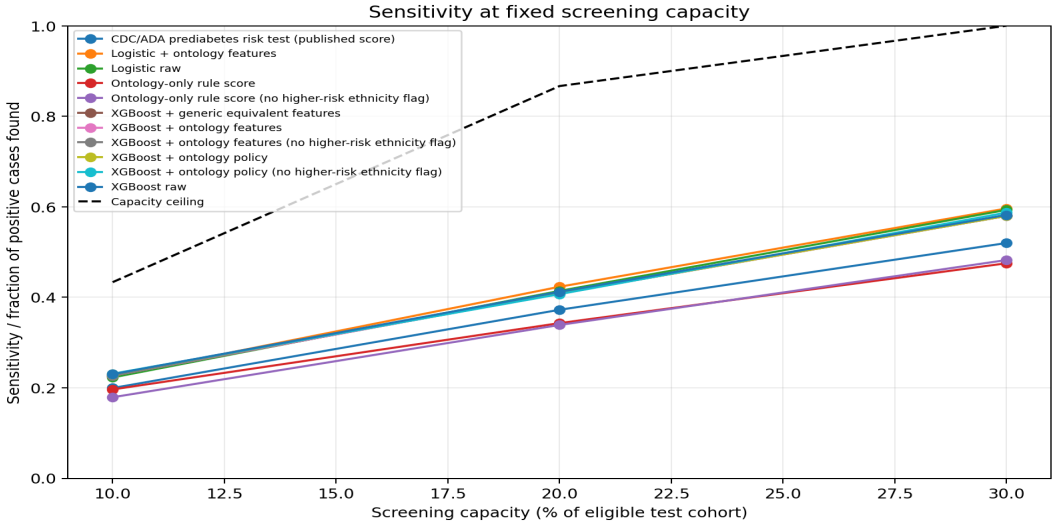


Figure 2: *Screening sensitivity across capacities (10%, 20%, 30%) relative to the prevalence ceiling.*

6.4 Primary Endpoint (H2, Confirmatory) — Null

The primary confirmatory comparison yielded a null result: hybrid sensitivity was 0.409 versus 0.412 for XGBoost alone. In the prespecified direction (hybrid minus ML-only), the difference was –**0.0031 (95% CI –0.0100 to +0.0079)**, lying entirely within the smallest detectable difference (0.0143).

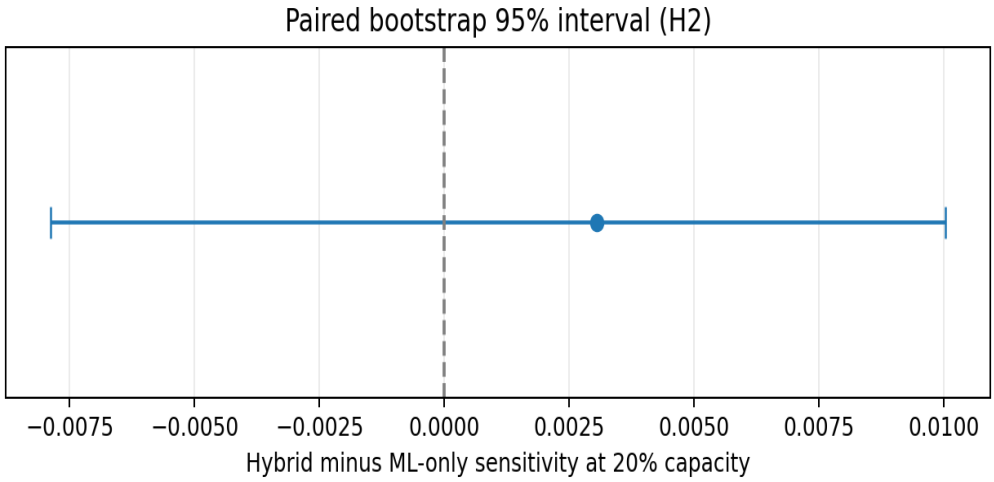


Figure 3: *Paired bootstrap confidence interval for the primary effect (hybrid vs ML sensitivity difference).*

6.5 Model-Family Knowledge-Layer Deltas (E3)

Adding ontology features to logistic regression yielded a sensitivity delta of +0.0092 (95% CI –0.0038 to +0.0185), while for XGBoost the delta was –0.0031 (95% CI –0.0181 to +0.0114). All intervals cross zero.

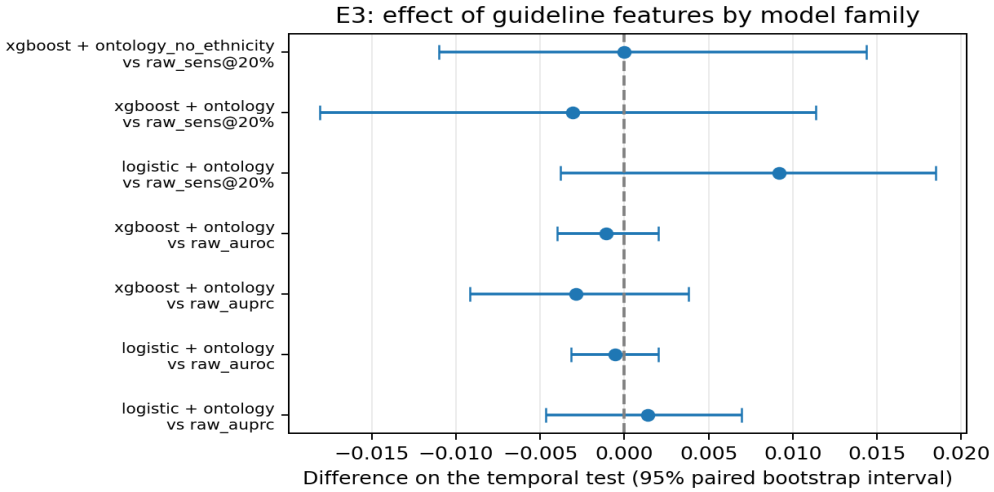


Figure 4: *Model-family knowledge-layer deltas across logistic regression and XGBoost.*

6.6 Low-Data Learning Curve (E1)

Evaluating models across training data fractions (5% to 100%) revealed no consistent low-data advantage for guideline features.

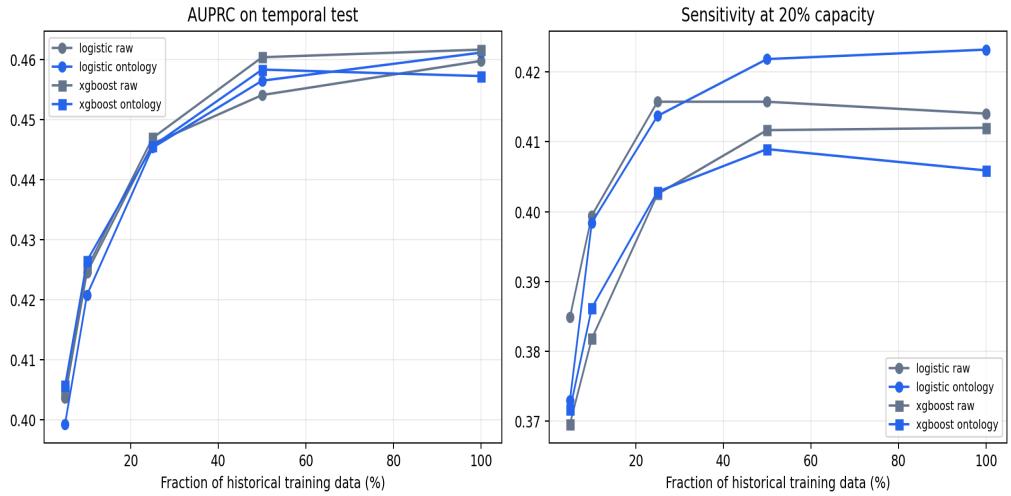


Figure 5: *Low-data learning curve showing performance deltas across training data fractions.*

6.7 Missing-Input Stress Test (E2)

Under 30% MCAR input masking of waist circumference, blood pressure, and hypertension history, the hybrid policy retained the highest AUPRC (0.4608) at every masking rate, and the linear model with ontology features exhibited the highest stability in capacity sensitivity.

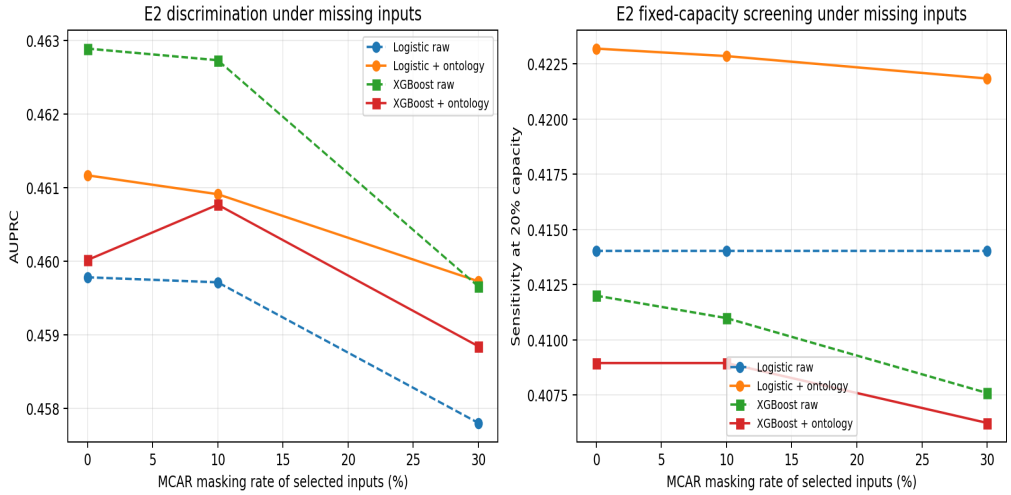


Figure 6: *Missing-input stress test showing robustness under MCAR data masking.*

6.8 Guideline Disagreement vs Confidence Review (H3) & Subgroups

Reviewing cases where the top-20% model set and rule set disagreed yielded a precision of 0.360 (750 participants), whereas reviewing participants closest to the ML cutoff achieved a precision of 0.400. Guideline disagreement was not enriched relative to confidence review.

7. Discussion

The primary screening endpoint is null, demonstrating that on rich tabular data with sufficient historical samples, explicit rule-derived features are largely redundant with raw model features for ranking. However, guideline integration provides substantial value in provenance tracking, auditability, and stability under missing inputs. The knowledge layer functions effectively as an explanation and governance substrate rather than a black-box

performance booster.

8. Limitations and Ethics

Study limitations include a cross-sectional single-measurement HbA1c outcome, unweighted NHANES complex survey design (preventing population-representative prevalence estimates), uncaptured screening criteria (family history, gestational diabetes), and the absence of prospective clinical validation.

9. Conclusion

Under a pre-temporally held-out evaluation on NHANES 2021–2023, adding an executable guideline-ontology layer produced a well-powered null primary effect on sensitivity at 20% capacity, while outperforming published point-score baselines and providing robust stability under missing data. The integration of formal knowledge models into tabular machine learning offers significant auditability and governance benefits while establishing rigorous performance boundaries.